

Vision and Scope Document

for

AI Study Hub

Prepared by Group 6

Process Impact

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5	Lê Hoài Sang	SE192192	Major Features; Swimlane Diagram
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1. Business Requirements

1.1. Background

Currently, FPT University students, especially those in the Software Engineering program, often have to manage various types of study materials such as lecture slides, textbooks, assignments, sample source code, past exam papers, and personal notes. However, these materials are often stored in a scattered manner across Google Drive, Messenger, Facebook Groups, email, or personal devices, making it difficult to search, categorize, and reuse them.

Furthermore, students lack a unified system for organizing materials by semester, subject, or topic. Sharing materials among students is still manual, leading to easy loss or duplication. In addition, students need to use AI to summarize materials, suggest classifications, and answer questions based on the content of the saved materials. Therefore, AI Study Hub is proposed as an online learning material management platform to support intelligent storage, searching, sharing, and learning for FPT University students.

1.2. Business Opportunity

To address the challenges of storing, searching, and using study materials, AI Study Hub provides a centralized platform to help students manage their materials more effectively. The system not only supports storing and categorizing materials by semester and subject but also integrates artificial intelligence (AI) to summarize content, suggest appropriate folders, and answer questions based on the user's study materials. This saves students time searching, improves self-study efficiency, and reduces reliance on decentralized storage channels like Zalo and Messenger.

1.3. Business Objectives

- **BO-1:** Reduce the average time students spend searching for study materials by 50% within 6 months of initial release.
 - **Scale:** Average time required to find a document (minutes)
 - **Metric:** User activity log and search history
 - **Past:** Average 10 minutes (internal survey)
 - **Goal:** Under 5 minutes
 - **Advanced Goal:** Under 3 minutes
- **BO-2:** Reduce the number of lost or inaccessible study materials by 80% within 6 months of initial release.
 - **Scale:** Percentage of lost or inaccessible materials reported by users
 - **Metric:** User feedback and incident reports
 - **Past:** Approximately 20-25% (Due to sharing habits via Zalo, large files are often deleted by the platform after a while).
 - **Goal:** Under 5%
 - **Advanced Goal:** Under 2%

1.4. Success Metrics

- **SM-1:** 40% of registered students use AI Study Hub at least twice per week within 6 months following initial release.
- **SM-2:** The average satisfaction rating for AI chatbot responses reaches 4.2 out of 5 within 3 months following initial release and 4.5 out of 5 within 12 months.
- **SM-3:** 80% of users report that AI Study Hub improves their ability to organize and retrieve learning materials within 6 months following initial release.
- **SM-4:** At least 90% of document search requests return relevant results within 3 seconds during the first 6 months following initial release.

1.5. Vision Statement

For students who struggle with fragmented study materials and limited accessibility, the AI Study Hub is a cloud-based and AI-integrated web application that will centralize document storage, enable intelligent semantic search, and provide a chatbot assistant for instant academic inquiry. Unlike the current manual process of searching through Google Drive, email, and various chat applications, students who use the AI Study Hub will have a single, organized source of truth for their learning materials, which will save significant time and increase the quality of their study sessions.

1.6. Business Risks

- **RI-1: API Cost Escalation:** The cost of using third-party LLM APIs could exceed the project budget if the number of requests scales rapidly beyond initial projections. (Probability = 0.5; Impact = 7)
- **RI-2: AI Hallucination/Inaccuracy:** The chatbot may provide incorrect information based on uploaded documents, potentially misleading students in their academic preparations. (Probability = 0.4; Impact = 9)
- **RI-3: Low User Adoption:** Students may continue to rely on traditional, familiar tools (e.g., Google Drive, Messenger), reducing the return on development effort and infrastructure costs. (Probability = 0.3; Impact = 8)
- **RI-4: Data Security Breach:** Unauthorized access to proprietary or private student documents could lead to privacy violations and loss of institutional trust. (Probability = 0.2; Impact = 10)

1.7. Business Assumptions and Dependencies

- **AS-1:** The selected cloud storage infrastructure will provide sufficient bandwidth and storage capacity to handle the expected volume of document uploads and real-time retrieval requests.
- **AS-2:** The AI models utilized will be capable of processing diverse file formats (PDF) with at least 90% accuracy in text extraction and contextual understanding.
- **AS-3:** Target users possess consistent high-speed internet access to interact with the web-based interface and AI engine.
- **DE-1:** The project is dependent on the availability and stability of third-party AI APIs (e.g., OpenAI or Google Gemini).

- **DE-2:** The application must integrate with a Vector Database (e.g., Pinecone or Milvus) to enable the RAG pipeline necessary for accurate document-based chatbot responses.

2. Scope and Limitations

2.1. Major Features

Module 1: Identity & Access Management (Core Authentication)

- **FE-01: Sign up** — Create a new student account using an email and password.
- **FE-02: Sign in** — Log in using registered email and password.
- **FE-03: Sign out** — Log out of the application and end the current session.
- **FE-04: Forgot password** — Recover accounts via email verification links.

Module 2: Document Management (PDF & Folders)

- **FE-01: Upload document (PDF Only)** — Upload study materials to the system; initial support is strictly limited to .pdf format.
- **FE-02: Document organization** — Organize uploaded files into user-created Folders and tag them under system-defined Semesters and Courses managed by the Admin.
- **FE-03: View document list** — Display a clean, page-by-page directory showing the user's personal folders and PDF files.
- **FE-04: Download document** — Download the original PDF files back to the local device.
- **FE-05: Edit document info** — Change file titles, descriptions, folder placement, and privacy configurations.
- **FE-06: Preview document** — Open and read the PDF content directly through an integrated online file viewer.
- **FE-07: Configure visibility** — Set file privacy to Private (only owner), Shared (specific links), or Public (everyone).
- **FE-08: Search document** — Using keywords or AI-supported semantic search.
- **FE-09: Delete document** — Allow users to delete their uploaded PDF files.

Module 3: Storage & File Handling

- **FE-01: Cloud upload** — Store documents on secure online servers with real-time status updates.
- **FE-02: File viewer integration** — Embed a dedicated document rendering component to ensure smooth online reading of PDF files within the browser.

Module 4: AI Chatbot (Document Q&A)

- **FE-01: Document Q&A** — Select PDF documents and open a chat interface to ask questions directly about their contents.
- **FE-02: Grounded responses** — Enforce system constraints so the AI chatbot generates responses based *only* on the facts inside the selected PDF files to prevent false information.
- **FE-03: Response stream tracking** — Display real-time text generation effects (streaming) alongside a loading indicator while the AI is computing an answer.
- **FE-04: Chat history retrieval** — Review past conversation logs saved and categorized by document.

Module 5: AI Quiz (Streamlined)

- **FE-01: Generate quiz** — Instruct the AI to scan a selected PDF document and automatically create a multiple-choice practice test.
- **FE-02: Score logging** — Provide an active test interface for users to submit answers; the system grades the test instantly and logs the final score and timestamp into their history.

Module 6: Public Library & Community Sharing

- **FE-01: Share to community** — Toggle a file to "Public" status to immediately publish it to the platform's global repository.
- **FE-02: Public library portal** — Provide a shared section displaying all public study materials from all users, allowing anyone to preview or download files.
- **FE-03: Secure link sharing** — Generate secure, direct web links (Secure URLs) to share private documents directly with chosen peers without publishing them to the public library.

Module 7: Admin Management (Core Infrastructure)

- **FE-01: Manage academic metadata** — A backend panel for administrators to seed, update, or remove Semester and Course categories so students can classify files properly.
- **FE-02: Content moderation** — Allow administrators to audit the global file directory and delete violating documents or suspend bad user accounts.

Module 8: Security & Privacy (Essential)

- **FE-01: Credential protection** — Secure all user passwords using a standard one-way digital cryptographic hashing function before saving them to the database.
- **FE-02: Logical data isolation** — Maintain a strict authorization barrier ensuring that files marked as Private are completely inaccessible to any other user or unauthorized AI query.

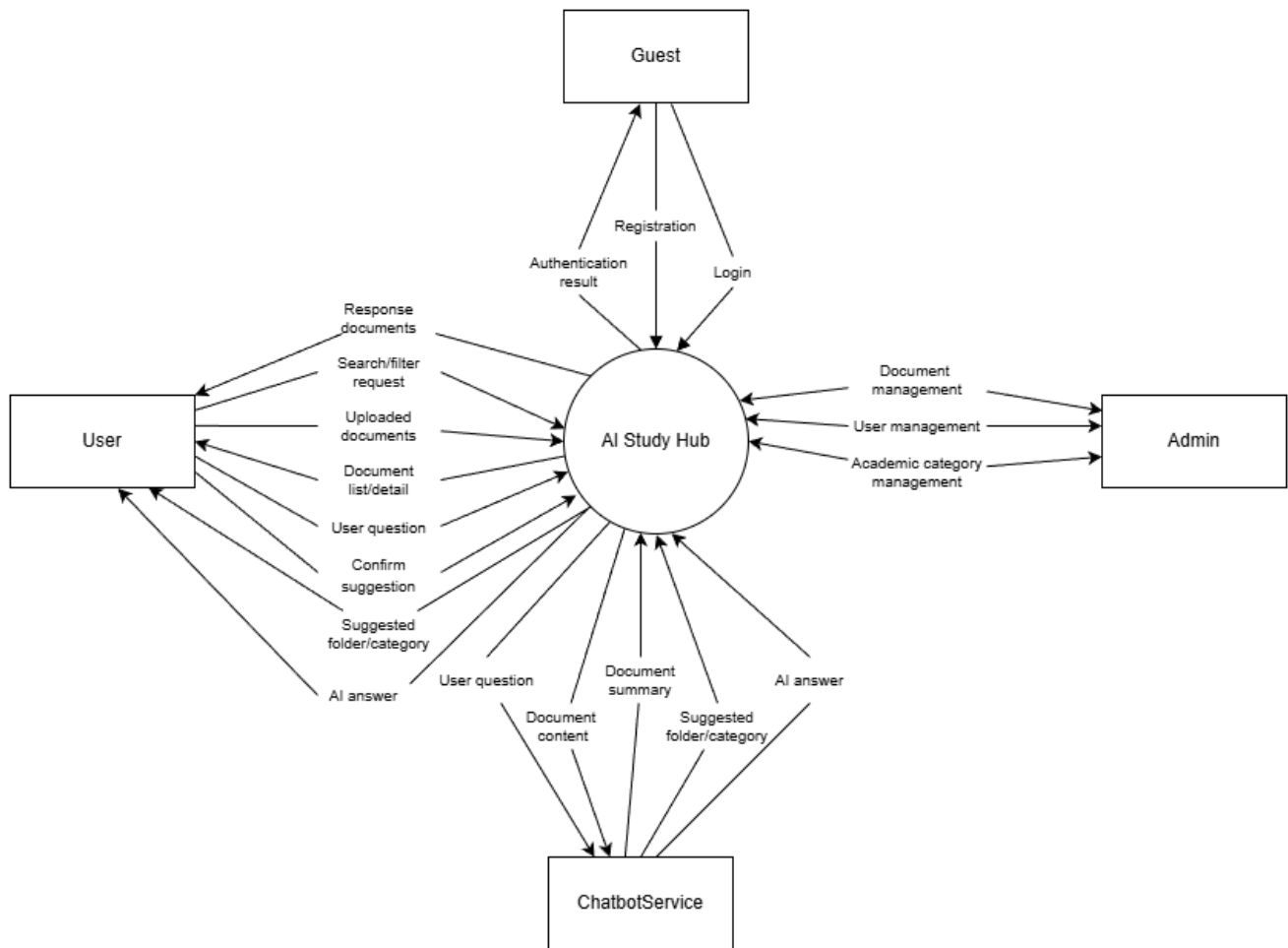


Figure 1. Context Diagram of AI Study Hub

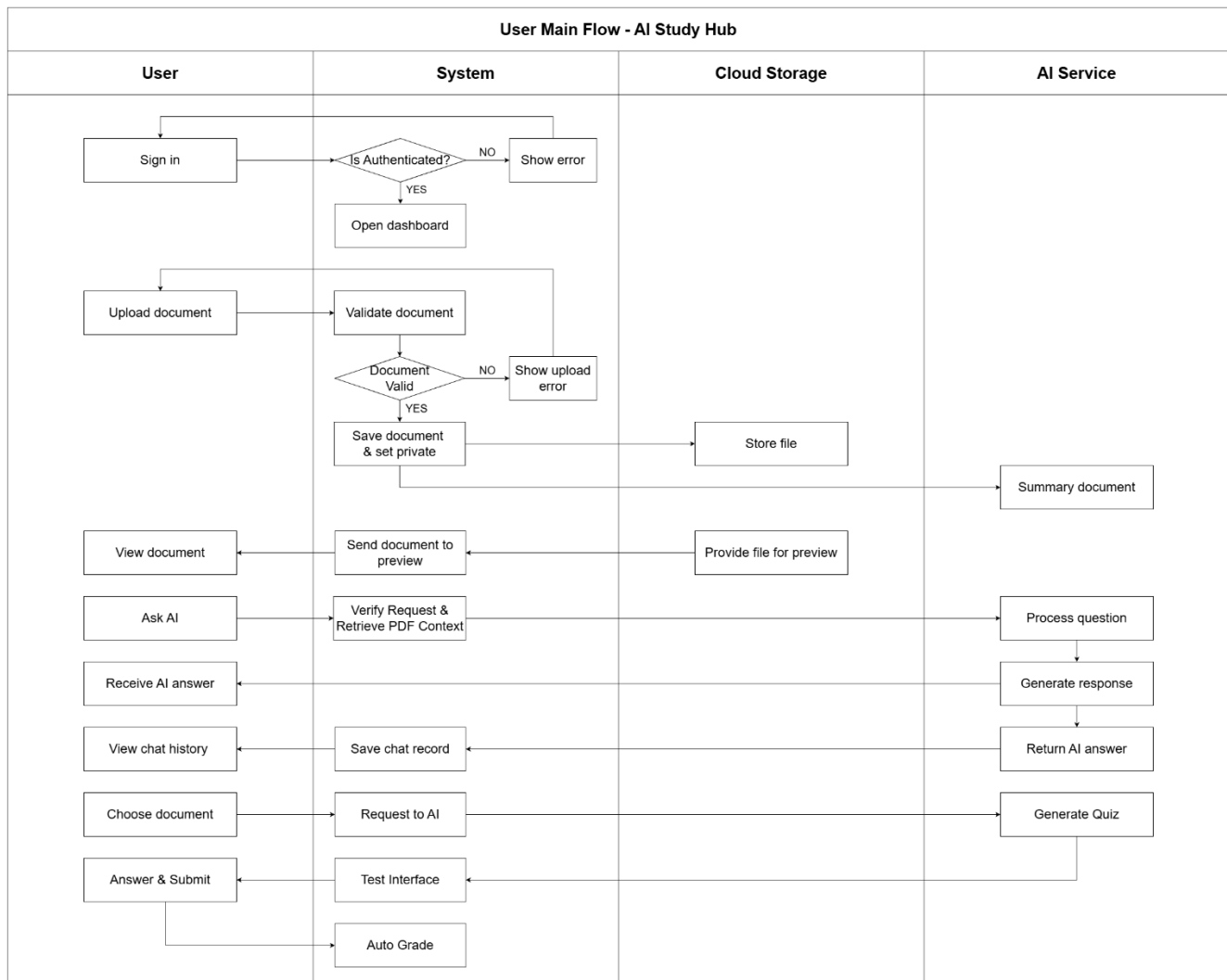


Figure 2.1. User Swimlane Diagram

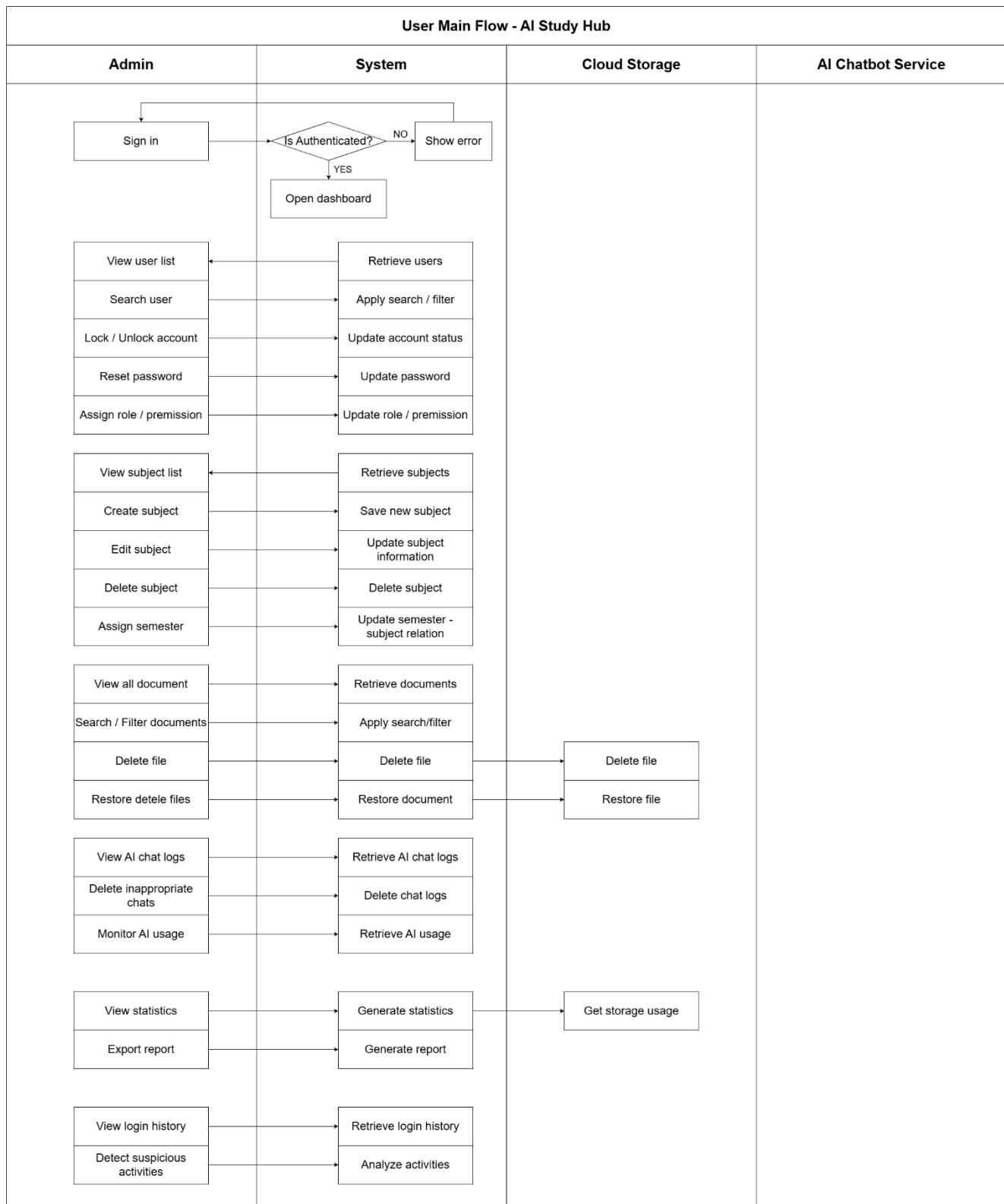


Figure 2.2. Admin Swimlane Diagram

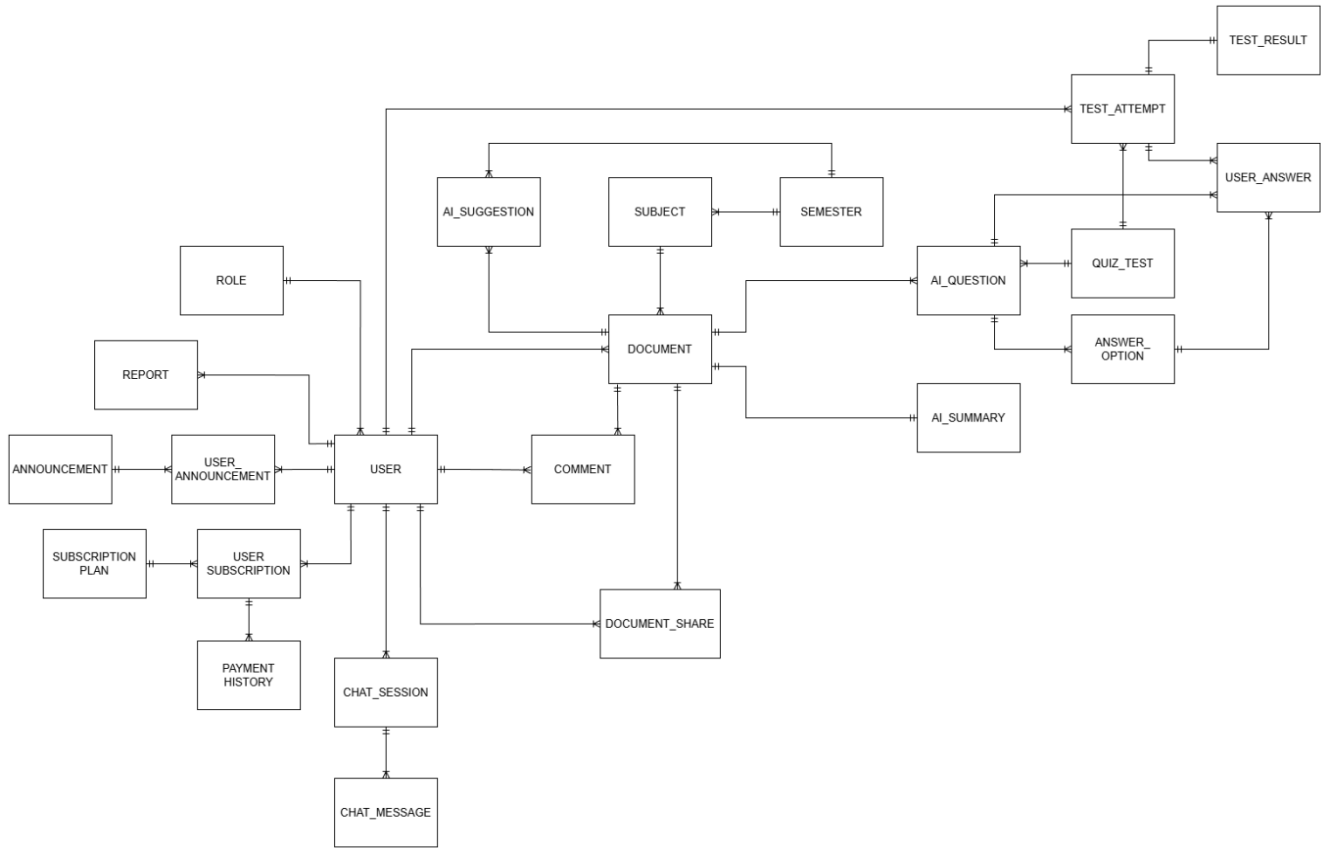


Figure 3. ERD

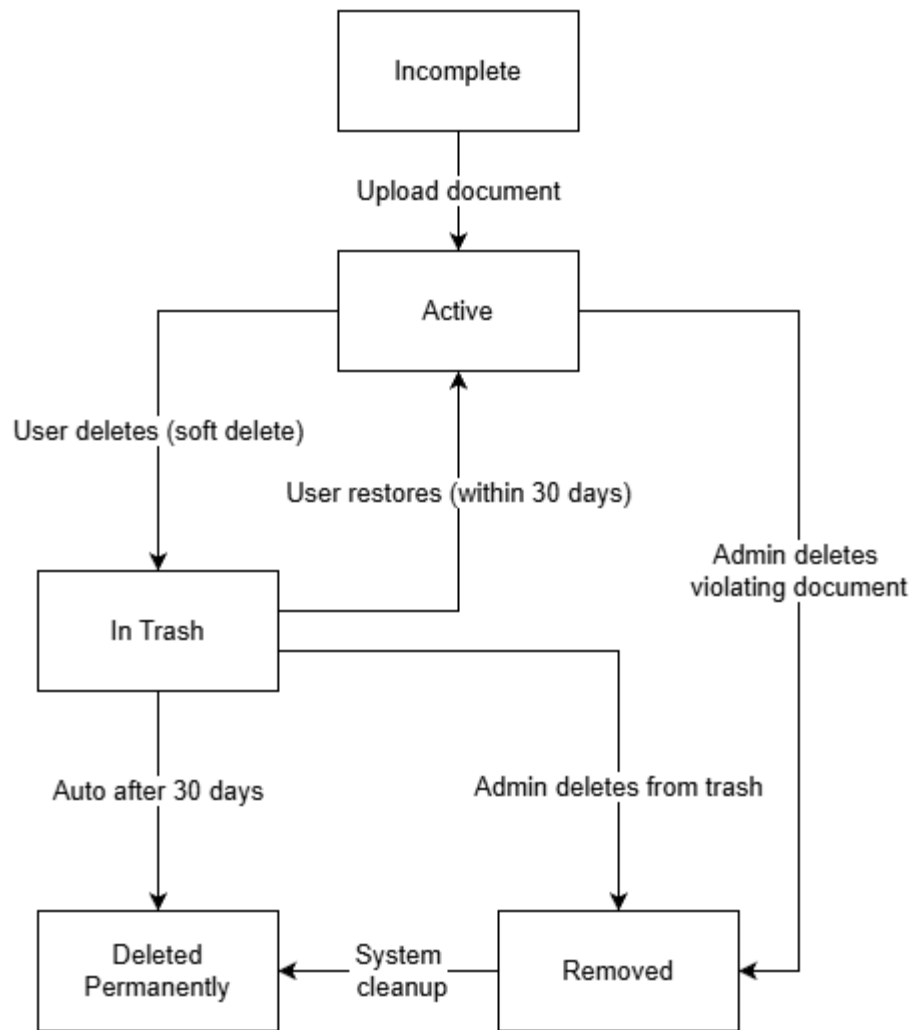


Figure 4. State-Transition of Documents



Figure 5. Use-Case Diagram of AI Study Hub

2.2. Scope of Initial and Subsequent Releases

Feature	Release 1	Release 2	Release 3
Identity & Access Management	- Sign up / Sign in - Basic profile display	- Active session directory & Remote logout	- Export learning logs (CSV/JSON) - Permanently delete chat/search history
Document Management	- PDF format only - Personal Folder-based organization - Assign files to Admin-defined - Semesters/Courses	- Support for DOCX, PPTX, and Images - "Shared with me" directory	- AI-driven folder suggestions based on document contents
Visibility & Sharing	- Visibility modes: Private and Public - Secure URL generation for quick private sharing	- Targeted sharing by inputting recipient emails (Grant View/Download access)	- Group-based Access Control Lists (ACL) for classes/study groups
AI Chatbot	- Contextual Q&A on a single document - Grounded responses to prevent false info	- Enforce daily message limits based on subscription tier	- Cross-document Q&A (Simultaneous chat across multiple files)
AI Quiz	- Automated multiple-choice quiz generation from 1 PDF - Timed test interface, auto-grading, and score logging	- Enforce monthly quiz generation limits based on subscription tier	- Export quizzes to PDF for offline printing and study
Cloud Storage	- Cloud file upload with real-time progress bar - Uncapped temporary storage for early adopters	- Enforce storage quotas (2GB/10GB) - Storage limit warnings and upgrade prompts	- Automated background file compression to optimize system space
Payment & Subscription	- Not applied (All features open for free to gather early usage data)	- Integrate payment gateway (PayOS/VNPay) - Automated plan provisioning (Standard/Pro tiers)	- Manage subscription renewals, refunds, and automated promo codes

Community Library	- Public library portal displaying shared PDF materials - Public online preview and file downloading	- Topic-based and semantic metadata filtering - Content flag/takedown button ("Report" button)	- AI-driven shared material recommendations based on user study habits
Admin Management	- Seed academic metadata (Semesters/Courses) - User directory oversight (Lock/Unlock accounts) - Manual removal of violating content	- Monitor transaction logs, order statuses, and system invoices	- Advanced Analytics Dashboard (Revenue trends, storage loads, token usage trackers, global rate limiting)

2.3. Limitations and Exclusions

- **LI-1: Third-party API Dependencies:** The system's operation (Cloud storage, AI Chat/Quiz, VNPay/PayOS payment gateway) is entirely dependent on APIs from third-party providers (such as Firebase, Gemini, banks). The project disclaims responsibility for service disruptions (downtime), delays, or policy changes from these providers.
- **LI-02: File Size:** Limits the maximum file size per upload (e.g., 50MB/file) to ensure system performance.
- **LI-03: Network Connected:** This system runs entirely on the cloud, requires a continuous internet connection, and does not support offline mode.

3. Business Context

3.1. Stakeholder Profiles

Stakeholder	Major Value	Attitudes	Major Interests	Constraints
Students / Users	Populate the platform with early user data. Drive system traffic and generate feedback. Share study materials to the library.	Eager to adopt tools that save study time. Sensitive to bugs or slow response times.	Centralized document storage. Instant and accurate AI document Q&A. Easy access to peer-shared materials.	Limited patience for complex user interfaces. Fully dependent on campus network/internet availability.
Student Contributors	Supply high-quality, structured learning materials. Build the initial value for the Public Library.	Willing to share knowledge but highly cautious about privacy and ownership.	Granular visibility controls (Private, Public, Secure Link). Clean and structured folder hierarchy.	High time and effort required to scan, compile, and clean up source documents.
System Administrators	Maintain backend stability, database backups, and user access security.	Highly risk-averse; prioritizes platform uptime, compliance, and budget control.	Data security, data isolation, and automated system monitoring.	Restricted by development timelines and infrastructure budgets (e.g., Cloud storage costs).
Development Team	Design UI/UX, program core RAG features, and optimize system infrastructure.	Focused on rapid feature delivery, code maintainability, and architectural stability.	Clean system architecture, error-free API integration, and stable deployments.	Limited development timeline (deadlines and fixed project milestones).
Third-party APIs	Provide Core AI models (Gemini/OpenAI) and cloud infrastructure.	Collaborative; seeks to validate model scalability through real-world user requests.	Stable API token consumption and low request latency.	Subject to sudden pricing structure updates or unexpected vendor service downtime.

3.2. Project Priorities:

Dimension	Constraint	Driver	Degree of Freedom
Schedule		Must be fully ready for the final project presentation defense panel by the end of this semester.	
Quality	95% of User Acceptance Tests (UAT) must pass. 100% of core security tests (data isolation between Private documents) must pass.		
Cost			A budget overrun of up to 15% is acceptable to handle unexpected AI token spikes during testing phases.
Features	All core features for Release 1.0 (including Semester Document Management, AI Chat Tutor) must be fully operational.		

3.3. Deployment Considerations

The AI Study Hub will be deployed as a responsive web system accessible via desktop and mobile browsers. It will feature a React frontend and a backend providing APIs for authentication, document management, AI Tutor interaction, practice test generation, and administration. A secure database and file storage will manage user accounts, courses, history, and uploaded materials (e.g., PDFs). Additionally, the system must strictly differentiate between public documents (accessible in the course library) and private documents (restricted to the owner).